Client Requirements

1. Client Overview

Client: Bokone Community Network (Vryburg)
Industry: Telecommunications
Client ID: CLI-148
Assigned Address Block: 172.30.100.0/23

Bokone Community Network is a telecommunications organisation based in Vryburg. Reliable internal connectivity is especially important in this environment, as technical staff depend on network access to monitor, manage, and support network operations. The network must therefore provide reliable connectivity, logical organisation, efficient use of resources, and capacity for future growth.

2. Reliable Network Connectivity

The network must allow authorised devices and segments to communicate successfully, verified through end-to-end testing rather than assumed from configuration. The design should minimise single points of failure and provide appropriate connectivity between important devices.

3. Logical Network Segmentation

The network will be logically separated by organisational function using VLANs and IP subnets, dividing the network into separate Layer 2 broadcast domains for scalability and traffic management. For Bokone, this separates functions such as:

- Administrative users
- Technical or network operations users
- Network management
- Servers and shared resources

For this implementation, technical/network operations and server traffic are combined into a single VLAN; the reasoning for this consolidation is detailed in the Logical Topology document

Since VLANs isolate broadcast domains, inter-VLAN communication requires Layer 3 routing (Cisco Systems, Inc., 2024). The proposed network includes an appropriate routing function to permit this communication while preserving segmentation. Exact VLAN and subnet allocation is defined in the Logical Topology and IP Addressing Plan.

4. IP Addressing Requirement

The 172.30.100.0/23 block falls within the private address space defined for internal networks (Internet Engineering Task Force, 1996) and forms the basis of the addressing plan, which must:

- Provide sufficient addresses for current network requirements
- Allocate appropriate subnets to each VLAN
- Identify network addresses, usable host ranges, and default gateways
- Minimise unnecessary address wastage
- Reserve capacity for future expansion

The full breakdown of each subnet is documented in the IP Addressing Plan.

5. Future Additional Floor

An additional floor may be occupied during the next financial year. The design reserves addressing capacity and a dedicated VLAN and subnet for future users and devices, detailed in the IP Addressing Plan, so that the network is not fully consumed by current use and does not require redesign later. The additional floor will not be physically implemented in the current Packet Tracer simulation; it is handled only through addressing and scalability design.

6. Future Branch Office (Change Request CR6)

A branch office is planned as part of the project change request. The project requires design and addressing accommodation only, with no second-site build. The network reserves appropriate addressing capacity and an extensible structure for when the branch office is eventually implemented, keeping the current build within the defined project scope.

7. Assigned Technical Challenge — EtherChannel

EtherChannel combines multiple physical Ethernet links into a single logical connection, providing load sharing and redundancy — traffic continues over the remaining links if one member link fails. This is appropriate for Bokone Community Network because an important switch-to-switch connection may carry traffic for multiple organisational segments; relying on a single physical uplink would create both a bandwidth limitation and a single point of failure. EtherChannel will therefore be implemented on that switch-to-switch connection (Cisco Systems, Inc., 2023).

The implementation must:

- Correctly configure the physical member interfaces
- Form the required logical EtherChannel
- Verify that the EtherChannel is operational
- Demonstrate connectivity across the aggregated link
- Test network behaviour when an individual member link is unavailable

Detailed configuration and verification will be presented in the Assigned Networking Feature / Technical Challenge section.

8. Scalability and Reliability Requirements

The overall design must:

- Use logical network segmentation
- Use the assigned 172.30.100.0/23 address block efficiently
- Provide reliable connections between important network devices
- Reserve a VLAN and subnet for the future additional floor
- Accommodate the future branch office through addressing and design only
- Allow additional devices and network segments to be introduced without unnecessary redesign

9. Testing and Verification Requirements

The completed network must demonstrate:

- End-to-end connectivity between appropriate devices
- Correct IP addressing and default gateways
- Communication between required VLANs
- Correct Layer 3 routing operation
- Correct operation of required network services
- Successful EtherChannel formation and operation
- Continued connectivity when an individual EtherChannel member link is taken offline

Any problems encountered during implementation will be investigated, resolved, and documented as part of the project's troubleshooting evidence.

10. Client Requirements Summary

Client Requirement	Design Response
Reliable internal connectivity	Structured topology with switching and L3 routing
Logical separation of functions	VLAN and subnet-based segmentation
Communication between VLANs	Layer 3 routing
Assigned addressing block	Efficient use of 172.30.100.0/23
Future additional floor	Reserved VLAN, subnet, expansion capacity
Future branch office	Addressing and design accommodation only
Reliable switch-to-switch link	EtherChannel
Network functionality	Correctly configured devices and services
Verification	End-to-end, routing, service, EtherChannel testing
Future scalability	Structure allowing growth without major redesign

11. Overall Requirement

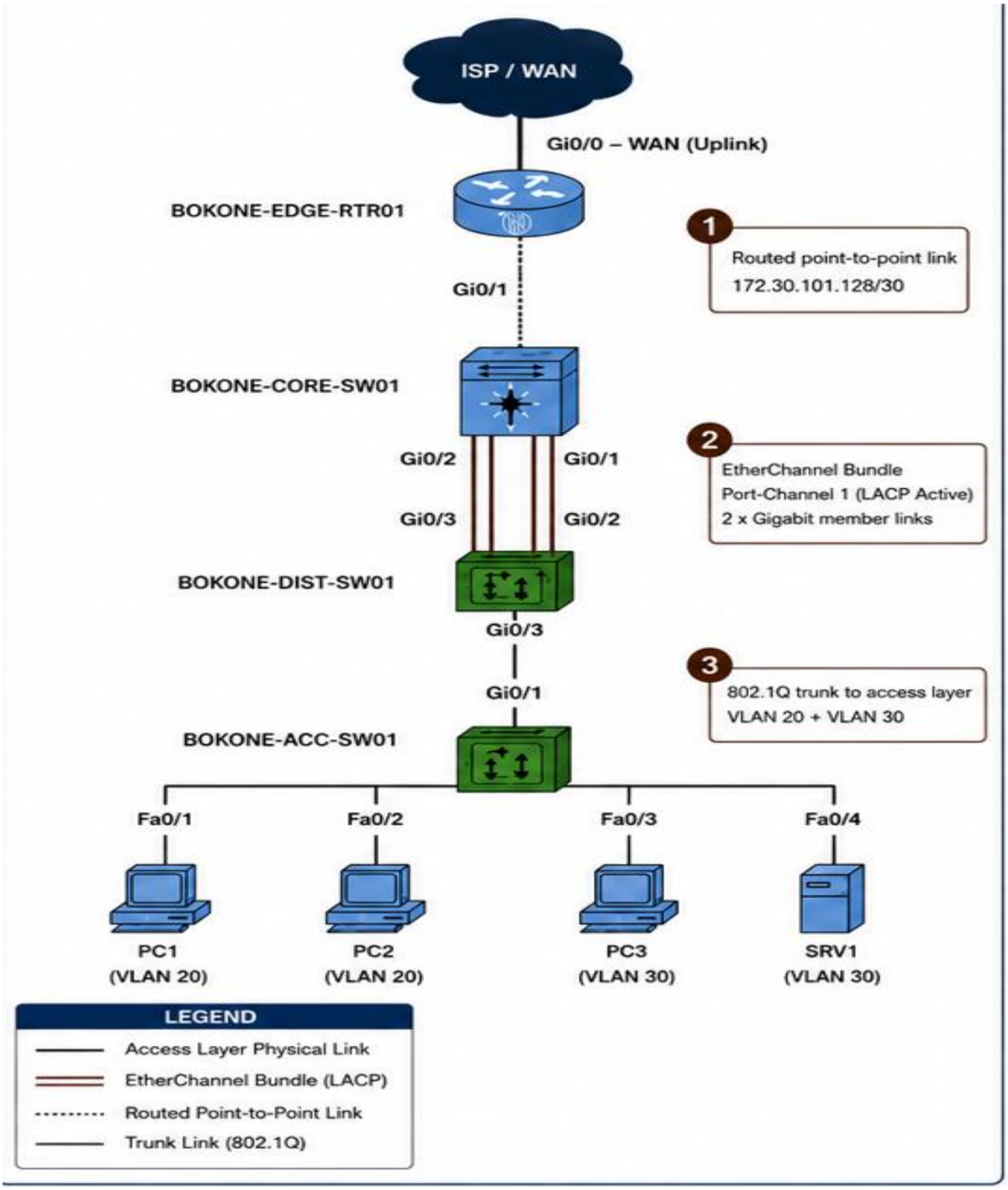
The proposed Bokone Community Network solution must be reliable, logically segmented, and scalable, using the assigned 172.30.100.0/23 address block efficiently to support the organisation's current connectivity requirements. VLANs and Layer 3 routing will organise and connect the network, while EtherChannel will provide a resilient switch-to-switch connection for important network traffic.

The network must also accommodate future growth by reserving a dedicated VLAN and subnet for the possible additional floor and providing addressing and design capacity for the planned branch office, without implementing a second site.

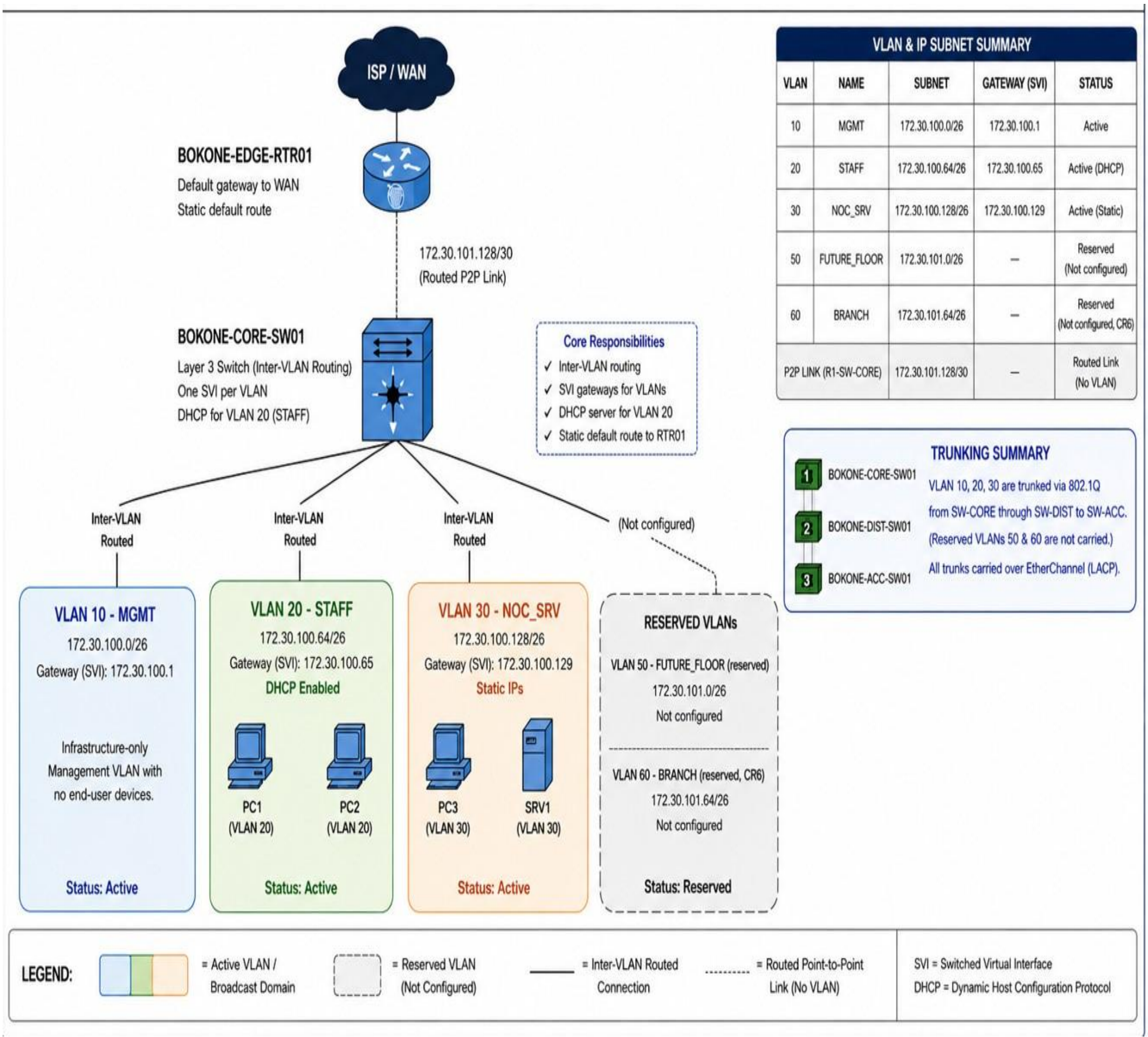
The complete solution will be designed, implemented, and tested in Cisco Packet Tracer, with all design decisions, configurations, testing results, and troubleshooting steps documented in the GitHub portfolio.

Topologies

1. Physical Topology



2. Logical Topology



IP Addressing Plan

All subnets are /26 unless noted. Full ranges shown as .x for brevity; complete addresses combine with the Network Address column.

VLAN	Name	Subnet	Network Addr	Broadcast Addr	Usable Range	Hosts	Gateway (SVI)	Status
10	MGMT	172.30.100.0/26	172.30.100.0	172.30.100.63	.1 – .62	62	172.30.100.1	Active (static)
20	STAFF	172.30.100.64/26	172.30.100.64	172.30.100.127	.65 – .126	62	172.30.100.65	Active (DHCP)
30	NOC_SRV	172.30.100.128/26	172.30.100.128	172.30.100.191	.129 – .190	62	172.30.100.129	Active (static)
—	Spare	172.30.100.192/26	172.30.100.192	172.30.100.255	.193 – .254	62	—	Unallocated (growth)
50	FUTURE_FLOOR	172.30.101.0/26	172.30.101.0	172.30.101.63	.1 – .62	62	—	Reserved
60	BRANCH	172.30.101.64/26	172.30.101.64	172.30.101.127	.65 – .126	62	—	Reserved (CR6)
—	P2P (R1–SW-CORE)	172.30.101.128/30	172.30.101.128	172.30.101.131	.129 – .130	2	—	Routed link
—	Spare	172.30.101.192/26	172.30.101.192	172.30.101.255	.193 – .254	62	—	Unallocated

GitHub Repository

GitHub Link [CMPG325-Bokone-Community-Network]

Reference

Cisco Systems, Inc. (2023) *Understand EtherChannel load balance and redundancy on Catalyst switches*. Available at: https://www.cisco.com/c/en/us/support/docs/lan-switching/etherchannel/12023-4.html (Accessed: 25 August 2026).

Cisco Systems, Inc. (2024) *Configure Inter-VLAN Routing with Catalyst Switches*. Available at: Cisco — Configure Inter-VLAN Routing with Catalyst Switches (Accessed: 25 August 2026).

Internet Engineering Task Force (1996) *RFC 1918: Address Allocation for Private Internets*. Available at: RFC 1918 — Address Allocation for Private Internets (Accessed: 26 August 2026).